

DAV Quiz 2 - A

Roll No: _____

Name: _____

Instructions: Do not write on the question paper or tick any option. Write selected option inside the brackets given at the end of each question "Answer []"

1. What model do traditional, unmanaged data lakes follow that makes them highly susceptible to inconsistency when write operations fail?

A) The ACID model (Atomicity, Consistency, Isolation, Durability). B) The relational model of normalized, third normal form schemas. C) The BASE model (Basically Available, Soft-state, Eventually consistent). D) The single-tier coupled compute and storage model. Answer []

2. To maintain platform flexibility and avoid "data silos," how should data teams conceptually view the Medallion architecture layers?

A) As physical servers isolated by hardware firewalls. B) As logical progressive refinement boundaries, rather than strictly rigid physical systems. C) As a single-tier coupled database schema where all tables share the same normal form. D) As a vendor-locked proprietary toolset native only to a single cloud provider.
Answer []

3. What is the primary functional intent of the Bronze layer in the Medallion framework?

A) To store raw data from diverse sources in its original structure, serving as a queryable, historical source of truth. B) To act as a highly aggregated, business-ready serving layer optimized for business intelligence tools. C) To completely normalize all incoming data into third normal form (3NF) to save on storage. D) To execute complex cross-domain joins and apply heavy business logic. Answer []

4. What unique metadata structure does Delta Lake establish alongside standard data files to track atomic transactions?

A) A separate on-premises SQL Server database metadata repository. B) The Hive Metastore database running inside a centralized Hadoop NameNode. C) A `_delta_log/` subdirectory containing ordered, sequential JSON commit files. D) An external schema registry mapped to static CSV files. Answer []

5. Why is premature cross-source integration (joining tables from different source systems) generally discouraged in the Silver layer?

A) Columnar file formats like Parquet cannot physically support tables with joined records. B) To maintain an isolated, decoupled design and prevent unnecessary dependencies between applications early in the pipeline.

C) Because joining tables automatically triggers the Delta VACUUM command and deletes historical snapshots. D) Joining data is exclusively reserved for unstructured generative AI vector databases.

Answer []

6. In the context of incremental data loads, what is the core functional difference between "append mode" and "merge mode"?

A) Append mode only adds new records to the end of the dataset; merge mode supports conditional upserts. B) Append mode deletes the existing table before writing; merge mode updates files in-place. C) Append mode relies on external database logs; merge mode is only compatible with unstructured files. D) There is no difference; both are synonyms for a complete truncate and reload of the table. Answer []

7. What is the functional role of an extra "landing zone" or "pre-Bronze" area in complex data platforms?

A) To act as a highly modeled dimensional presentation area for executive BI dashboards. B) To run computationally intensive machine learning model training passes directly on source files. C) To replace Delta Lake tables with volatile, temporary memory caches. D) To stage raw data, perform safe decompression, metadata generation, and protect the system against unexpected API errors. Answer []

8. What operational problem does Delta Lake's built-in OPTIMIZE command resolve?

A) The "schema-on-read" bottleneck caused by missing conformed dimensions. B) The storage cost of maintaining historical files by instantly deleting all time-travel snapshots. C) The physical file fragmentation and high network read overhead of the "small file problem." D) The lack of secure views for column-level data masking. Answer []

9. How does Delta Lake speed up transaction log reconstruction?

A) It periodically writes standard, consolidated "checkpoint" snapshots of the log, typically every 10 transactions. B) It translates the transaction log into highly normalized Data Vault tables. C) It restricts all table operations to a single, centralized on-premises server. D) It runs an asynchronous machine learning classification algorithm to predict future transactions. Answer []

10. Which of the following is NOT an advantage of a well-designed data architecture?

A. Reduced data redundancy and storage costs across systems B. Elimination of initial engineering setup costs and overhead C. Improved data governance, lineage tracking, and compliance D. Enhanced query performance and scalability for analytics Answer []

Answer

1. C
2. B
3. A
4. C
5. B
6. A
7. D
8. C
9. A
10. B